

QUARE: Quality-Aware Requirements Engineering through Multi-Agent Dialectical Negotiation

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Abstract—Requirements engineering (RE) is critical to software success, yet automating it remains challenging because multiple, often conflicting quality attributes must be balanced while preserving stakeholder intent. Existing Large-Language-Model (LLM) approaches predominantly rely on monolithic reasoning or implicit aggregation, limiting their ability to systematically surface and resolve cross-quality conflicts. We present QUARE (Quality-Aware Requirements Engineering), a multi-agent framework that formulates requirements analysis as structured negotiation among five quality-specialized agents (Safety, Efficiency, Green, Trustworthiness, and Responsibility), coordinated by a dedicated orchestrator. QUARE introduces a dialectical negotiation protocol that explicitly exposes inter-quality conflicts and resolves them through iterative proposal, critique, and synthesis. Negotiated outcomes are transformed into structurally sound KAOS goal models via topology validation and verified against industry standards through retrieval-augmented generation (RAG). We evaluate QUARE on five case studies drawn from established RE benchmarks (MARE, iReDev) and an industrial autonomous-driving specification, spanning safety-critical, financial, and information-system domains. Results show that QUARE achieves 98.2% compliance coverage (+105% over both baselines), 94.9% semantic preservation (+2.3 percentage points over the best baseline), and high verifiability (4.96/5.0), while generating 25–43% more requirements than existing multi-agent RE frameworks. These findings suggest that effective RE automation depends less on model scale than on principled architectural decomposition, explicit interaction protocols, and automated verification.

Index Terms—Requirements Engineering, Multi-Agent Systems, Large Language Models, KAOS Modeling, Goal-Oriented Requirements Engineering, Requirements Negotiation

I. INTRODUCTION

Requirements Engineering (RE) serves as the foundational phase of the software development lifecycle, directly influencing the quality and success of the final product [1]. In the era of autonomous systems and AI-driven applications, RE has evolved from simple functional specifications to encompass a multi-dimensional spectrum of quality attributes [2]. Modern software must simultaneously satisfy competing non-functional requirements (NFRs), such as safety, efficiency, environmental sustainability, and trustworthiness [3]. As system

complexity scales, manually balancing these diverse quality dimensions while maintaining consistency becomes error-prone and labor-intensive [4].

Despite decades of research, RE remains a primary source of software project failure, with empirical studies indicating that over 70% of failed projects trace back to requirements-related issues such as incompleteness, ambiguity, and unresolved stakeholder conflicts [5], [6]. Safety-critical domains such as autonomous vehicles and medical devices amplify these challenges, as systems must simultaneously satisfy stringent, often competing, quality attributes [7]. The core difficulties include balancing multi-dimensional quality attributes, resolving inconsistencies across heterogeneous stakeholder requirements, and providing systematic modeling support with sufficient traceability for downstream validation.

These challenges persist due to structural limitations of existing RE approaches. First, requirements analysis is typically performed from a monolithic perspective, making it difficult to reason explicitly about conflicts. Both human-driven and single-agent automated approaches tend to collapse diverse stakeholder viewpoints into a unified narrative, obscuring underlying trade-offs. Second, while goal-oriented modeling methods such as KAOS (Knowledge Acquisition in Automated Specification) [8] provide expressive formalisms, they remain largely manual. Third, most existing approaches lack explicit negotiation protocols. We term this the *Monolithic Integration Gap*, i.e., the absence of structured dialogical mechanisms through which agents with competing quality objectives can surface, categorize, and resolve conflicts. Without a protocol that distinguishes resource-bound conflicts from logical incompatibilities, multi-agent systems default to implicit aggregation, which obscures the rationale for trade-offs.

To address these limitations, we propose QUARE, a quality-oriented multi-agent framework grounded in dialogical negotiation. The core innovation lies in decomposing requirements analysis along critical quality dimensions derived from ISO/IEC 25010 [3] and assigning each to a specialization based on a Large Language Model (LLM). Rather than em-

playing simple voting or sequential pipelines, these agents engage in explicit, multi-round negotiation through a five-phase pipeline that identifies and resolves cross-quality conflicts. QUARE then synthesizes negotiated outcomes into standardized KAOS goal models, ensuring structural correctness and industry compatibility.

The main contributions of this work are as follows:

- A **negotiation-based multi-agent architecture** for RE that formulates requirements analysis as collaborative negotiation among quality-specialized agents with explicit conflict resolution across quality dimensions.
- An **automated pipeline for KAOS goal modeling** that transforms unstructured requirements into structurally valid, traceable goal models.
- A **geometry-based evaluation methodology** for multi-quality RE that addresses the limitation of scalar metrics by operationalizing requirement diversity as convex hull volume (CHV) and mean distance to centroid (MDC) in a five-dimensional quality space, complemented by per-axis balance metrics (Coverage Uniformity and Minimum Axis Coverage), phase-wise semantic drift analysis (BERTScore), and structural compliance verification.
- An **open-source reimplementation suite (Open-ReBench)**¹ containing faithful reimplementations of all compared frameworks under identical conditions, necessitated by the absence of public implementations of MARE and iReDev, along with all experimental artifacts, prompts, outputs, and evaluation scripts.

II. RELATED WORK

Early research on RE has long explored automating requirements analysis. For example, Vinay et al. [9] proposed a rule to automate requirements analysis and partial elements analysis. However, such early methods relied heavily on predefined heuristics and lacked mechanisms to reason about conflicting quality attributes or evolving stakeholder intent.

A. LLM-based Requirements Engineering

Recent advances in LLMs have stimulated growing interest in automating RE tasks, including requirements elicitation, classification, refinement, and specification generation [10], [11]. Elicitron [12] employs a population of LLM-generated user agents to simulate diverse perspectives and uncover latent needs; however, these agents operate independently without inter-agent negotiation.

Despite promising results, most LLM-based RE approaches rely on single-agent reasoning [13], [14], where a single model performs all analysis within a unified prompt. Such approaches implicitly optimize for linguistic plausibility rather than multi-objective trade-offs. When multiple quality attributes are encoded together, dominant objectives tend to overshadow less explicit concerns such as sustainability or regulatory responsibility. Moreover, these methods provide limited support for explicit conflict identification, traceability across artifacts, or semantic guarantees for downstream verification.

B. Multi-Agent Systems in Requirements Engineering

Collaborative RE has sought multi-agent systems. Early work such as Xipho [15] and the negotiation-based frameworks of Grünbacher et al. [16] model requirements analysis as interactions among autonomous agents, demonstrating that negotiation can surface conflicts and improve consistency.

More recently, the literature has proposed LLM-based multi-agent frameworks to automate larger portions of the RE lifecycle. MARE [17] decomposes RE into elicitation, modeling, verification, and specification tasks, with task-specialized agents collaborating through a shared workspace. iReDev [18], [19] augments this with knowledge-driven agents and a human-in-the-loop mechanism for artifact refinement. Both frameworks demonstrate the feasibility of end-to-end automation but primarily emphasize task orchestration rather than quality-dimension negotiation. Specifically, the agents in MARE share the same holistic quality perspective and lack the structural tension necessary to surface cross-quality conflicts. iReDev resolves conflicts through external human intervention or implicit convergence, without a protocol that categorizes conflict types or enforces semantic preservation guarantees. Neither framework explicitly models negotiation dynamics among competing quality attributes, leaving the rationale behind trade-off decisions opaque.

C. Argumentation Theory and Negotiation

Dialogical approaches and negotiation have roots in formal argumentation theory. The abstract argumentation framework of Dung [20] formalized argument defeat and acceptability, while Walton and Krabbe [21] characterized dialogue types, including persuasion and negotiation. The protocol of QUARE most closely resembles a persuasion dialogue combined with a negotiation dialogue: agents propose requirements grounded in quality objectives (thesis), peers articulate critiques that expose constraint violations (antithesis), and a moderator synthesizes refined proposals (synthesis).

Recent work has explored LLM-based multi-agent debate for improving factual accuracy [22], [23], demonstrating that structured disagreement improves output quality beyond single-agent self-critique. However, these approaches target open-ended reasoning tasks and do not address RE-specific challenges: conflict categorization by quality dimension, termination under engineering feasibility constraints, and semantic preservation of stakeholder intent.

QUARE differentiates itself by grounding agent roles in a domain-specific quality taxonomy (ISO/IEC 25010) rather than generic debate roles, incorporating explicit conflict classification (resource-bound vs. logical incompatibility) with priority-weighted resolution. Convergence is evaluated via semantic stability (BERTScore [24]) and engineering-level consistency criteria (ISO/IEC/IEEE 29148, DAG validity), so that outputs are both linguistically coherent and engineering-ready.

To our knowledge, QUARE is the first framework in the RE domain to systematically instantiate an argumentation-inspired dialectical negotiation protocol within an LLM-based pipeline,

¹<https://anonymous.4open.science/r/OpenRE-Bench-6660>

TABLE I: Specialized Agents within QUARE

Agent Role	Quality Concern	Core Responsibility
Safety Agent	Safety & Reliability	Hazard identification and mitigation
Efficiency Agent	Performance	Resource optimization and latency
Green Agent	Sustainability	Energy efficiency and carbon footprint
Trustworthiness Agent	Security & Privacy	Data protection and access control
Responsibility Agent	Ethical & Compliance	Regulatory and social responsibility
Orchestrator Agent	System Integration	Multi-agent coordination and workflow orchestration

featuring explicit conflict categorization and engineering-level convergence criteria.

III. METHODOLOGY

The following research questions (RQs) guided the research in our work:

RQ1: Does specialized multi-agent collaboration improve the comprehensiveness and quality of generated requirements compared to single-agent generation?

RQ2: How does multi-agent negotiation affect conflict resolution effectiveness while preserving the original semantic intent of requirements?

RQ3: How effectively does the framework generate structurally valid and industry-compliant requirement models?

A. Multi-Agent Architecture

To address the multi-dimensional nature of requirements quality, we adopt a role-based multi-agent architecture comprising six distinct roles: five quality-specialized agents and one orchestration role (Table I). The quality-specialized agents represent core non-functional concerns: Safety, Efficiency, Sustainability, Trustworthiness, and Responsibility, while a dedicated Orchestrator coordinates these agents and integrates their outputs.

The five specialized agents are instantiated using a shared LLM backbone but are functionally isolated through dedicated system prompts. Each prompt comprises three components: (1) a role definition specifying the quality dimension and prioritization criteria; (2) a chain-of-thought (CoT) task instruction directing the agent to analyze input, identify quality-relevant concerns, and formulate requirements; and (3) an output schema enforcing JSON-structured requirements with fields for goal ID, description, quality dimension, KAOS level, and rationale. All agents use zero-shot prompting.

The selection of these five quality dimensions is grounded in the ISO/IEC 25010 SQuaRE (Systems and Software Quality Requirements and Evaluation) quality model [25], which defines a hierarchical taxonomy of eight product quality characteristics. Safety and Reliability correspond to the Reliability and Safety characteristics of ISO 25010; Efficiency maps to Performance Efficiency; Trustworthiness subsumes the Security characteristic; and Responsibility captures Maintainability and compliance-oriented aspects formalized in recent

extensions for AI systems [26]. The Sustainability (Green) dimension, while not explicitly present in the original standard, reflects the growing consensus on green software [27] and has been proposed as an extension in recent literature [28]. This grounding ensures principled, non-overlapping agent decomposition with established measurement criteria, in contrast to MARE and iReDev, which do not ground their decompositions in an explicit quality standard.

The rationale for this specialization is twofold. First, it mitigates prompt overloading, where an LLM may overlook attribute-specific constraints when exposed to an overly broad prompt [28], [29]. Second, it deliberately induces structured disagreement among agents with competing objectives (e.g., Safety vs. Efficiency), thereby laying the groundwork for the subsequent dialogical negotiation phase (Section III-C).

B. The Five-Phase Requirement Pipeline

The operational workflow is organized into a five-phase pipeline (Figure 1), ensuring systematic transition from raw project descriptions to formalized goal models. The pipeline separates concerns across phases: diversity-oriented generation (Phase 1), semantic reconciliation via negotiation (Phase 2), structural consolidation through model integration (Phase 3), constraint enforcement through verification (Phase 4), and final materialization (Phase 5). Negotiation and verification serve distinct roles: conflicts are resolved at the semantic level before structural integration. At the same time, compliance checking is enforced as an independent validation gate that does not alter negotiated intent.

Phase 1: Parallel Generation

Five specialized agents operate in parallel, each processing the same project description using quality-oriented system prompts, producing diverse candidate requirements that cover functional and non-functional concerns.

Phase 2: Dialectical Negotiation

The coordinator identifies semantic overlaps and logical contradictions. Agents then engage in multi-round deliberation following a thesis, antithesis, and synthesis structure to refine or merge conflicting goals. Each round concludes with pairwise cross-agent validation. The negotiation protocol is detailed in Section III-C.

Phase 3: Integration & Topology Validation

Fragmented requirements are synthesized into a coherent KAOS structure through semantic deduplication, cross-agent parent-child stitching, and DAG validation. Cycles are prohibited as they would imply that a goal depends on its own sub-goals, creating circular reasoning that undermines traceability. Structural corrections preserve negotiated intent from Phase 2.

Phase 4: Verification

Multi-layer validation includes deterministic rule checking, RAG-based hallucination detection, and compliance verification (e.g., ISO 26262, ISO 27001). This phase augments requirements with structural annotations without modifying underlying intent.

Phase 5: Standardized Output Generation

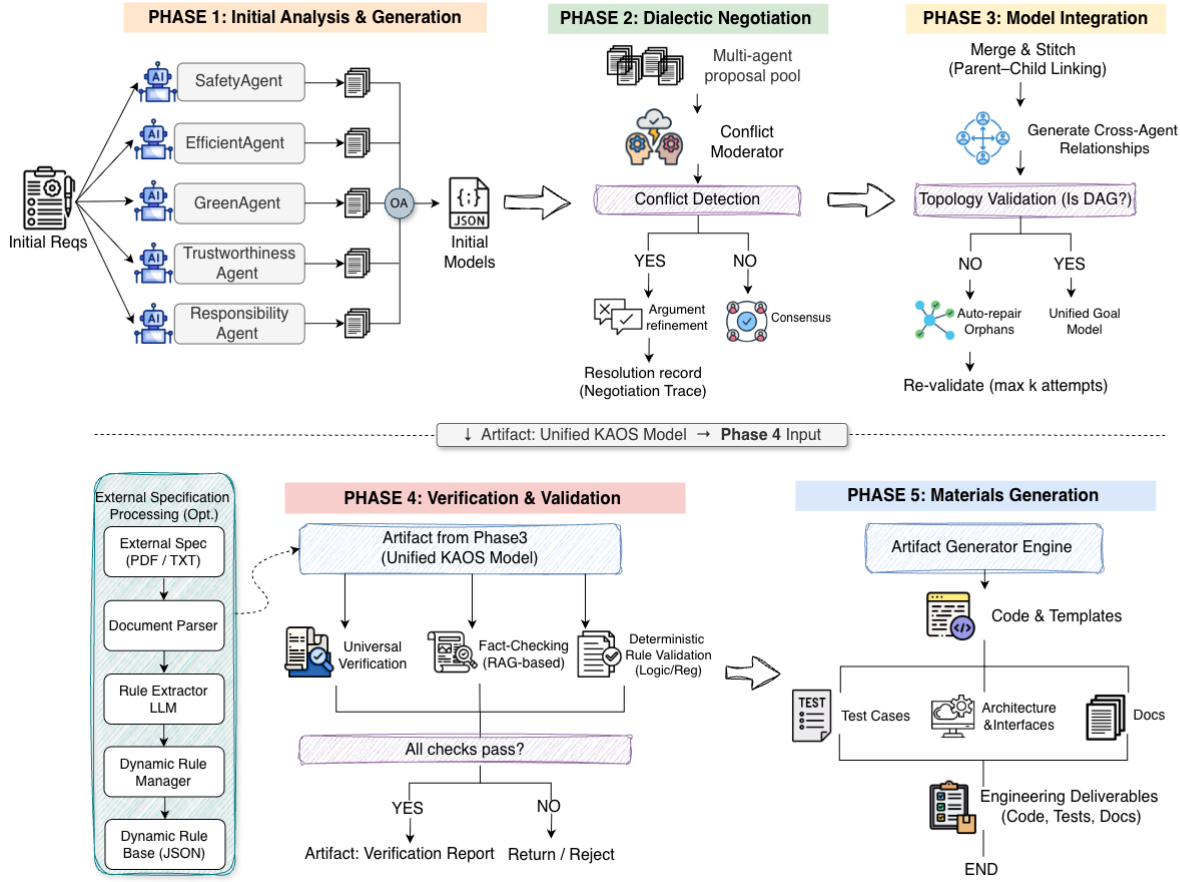


Fig. 1: Overview of the QUARE framework. The pipeline comprises five sequential phases: parallel agent-based generation (Phase 1), dialectical conflict negotiation (Phase 2), KAOS goal model integration and topology validation (Phase 3), multi-layer verification and compliance checking (Phase 4), and standardized engineering materials generation (Phase 5).

Generates formalized KAOS models (JSON and GSN XML), human-readable documentation, and downstream materials, such as test cases and architecture documents.

C. Dialectical Negotiation Protocol

The protocol instantiates a structured dialectical negotiation inspired by formal argumentation [20], adapted to multi-quality requirements negotiation. We adopt *dialectical* as the primary term to emphasize the thesis–antithesis–synthesis structure. The protocol is also *dialogical* in the sense of Walton and Krabbe [21], as it unfolds through multi-party dialogue. Each round consists of a thesis (agent proposal), antithesis (peer critique citing constraint violations), and synthesis (moderated resolution), with explicit engineering termination and traceability conditions. Unlike implicit aggregation or heuristic-based debate, the protocol treats conflict resolution as a primary operation, prioritizing semantic preservation alongside transparent trade-off management.

a) Conflict Detection and Categorization: We define a conflict as a state in which the satisfaction of an agent-specific constraint C_i undermines the attainment of another constraint C_j . Conflicts fall into two classes. The first concerns *resource-bound conflicts*, which arise when quality attributes compete

for finite resources (e.g., safety redundancy vs. power minimization). The second consists of *logical incompatibilities*, in which requirements impose mutually exclusive system states (e.g., encryption latency vs. response-time constraints).

A specialized Conflict Coordinator mediates conflict detection in two stages. First, the coordinator computes pairwise cosine similarities over `bert-base-uncased` embeddings of agent-generated requirements and flags pairs whose similarity exceeds $\tau = 0.85$ as potential overlaps. Second, the LLM classifies each flagged pair as (a) redundant, (b) resource-bound conflict, or (c) logical incompatibility. Separating embedding-level retrieval from LLM-level classification allows the coordinator to surface multi-dimensional contradictions that heuristic merging would absorb.

b) Dialectical Loop: The protocol operates through three coordinated stages. In the first stage, agents independently analyze requirement pairs to detect semantic overlaps and logical contradictions, establishing a conflict registry that prioritizes high-severity incompatibilities. The second stage initiates dialectical rounds, in which agents engage in iterative proposal, critique, and synthesis cycles, capped at 3 rounds to ensure termination. The third stage validates the resolution outcome, verifying that all registered conflicts have been addressed and

that no new contradictions have been introduced.

In each iteration, a focus agent A_i proposes a candidate requirement set R_i reflecting its primary quality concern. Peer agents evaluate R_i against their own constraints and articulate rationalized critiques. These critiques are integrated through neutral moderation, resolving conflicts either by jointly satisfying multiple constraints or by applying project-specific priority weights. Unresolved conflicts after three rounds are escalated to priority-weighted integration in Phase 3, preserving all conflict rationale in the negotiation trace.

By design, the negotiation phase records conflict signals, resolution states, and trade-off rationale as structured metadata rather than directly rewriting requirement text. Textual modifications are deferred to Phase 3, where deduplication and structural harmonization are applied.

c) Mathematical Formalization: Let $\mathcal{A} = \{A_1, \dots, A_n\}$ denote specialized agents with quality objectives Q_i and priority weights w_i . The negotiation approximates:

$$r^* = \arg \max_r \sum_{i=1}^n w_i \cdot Q_i(r) \quad (1)$$

The weights w_i are initialized uniformly ($w_i = \frac{1}{n}$) and can be overridden by project-specific configuration. At round k , requirement r_k is updated via:

$$r_{k+1} = G(r_k, \text{critique}(A_j, r_k)) \quad (2)$$

where G denotes the synthesis function integrating peer critiques into a revised proposal, the loop terminates when semantic similarity between successive rounds exceeds $1 - \epsilon$ (measured by BERTScore) or $N = 3$ rounds is reached, ensuring both convergence and bounded computation.

Eq. (2) is, in principle, order-sensitive. To mitigate ordering bias, QUARE employs round-robin scheduling in which each agent serves as the focus agent exactly once per round, and the Conflict Coordinator aggregates all pairwise critiques before synthesis. While this does not guarantee order-invariance, experiments across three random seeds (which shuffle agent ordering) show low variance, suggesting that the multi-round structure and coordinator-mediated synthesis substantially reduce order sensitivity in practice.

IV. EXPERIMENTAL SETUP

A. Case Studies

We evaluate QUARE using five case studies spanning safety-critical, financial, and information system domains (Table II). The Library and ATM systems are drawn from the MARE benchmark [17], while the RollCall and Bookkeeping systems are adopted from iReDev [18]. We additionally include an Autonomous Driving (AD) case derived from industrial specifications [30], representing a safety-critical system with rich cross-quality constraints. Across all five case studies, specialized agents are activated to ensure comprehensive quality coverage and enable fair comparisons across systems with varying quality emphases.

TABLE II: Summary of Case Studies

Case Study	Domain	Core Focus
AD	Automotive	Safety (ISO 26262)
Library System	Management	Scalability
ATM System	Finance	Security / Compliance
RollCall System	Management	Functional Logic
Bookkeeping System	Finance	Data Integrity

B. Agent and Model Configuration

The implementation allows model alternation, and the experiment used only the `gpt-4o-mini-2024-07-18` model. The temperature was fixed at 0.7 to balance linguistic diversity and structural consistency. This lightweight model configuration isolates the impact of the proposed architecture and negotiation mechanisms from model scale effects. Agents employ ChromaDB for retrieval-augmented generation (RAG) [31], grounding their reasoning in external standards including ISO 26262 and ISO 27001. Generated goal models conform to a three-level KAOS hierarchy (Strategic, Tactical, and Operational). The negotiation process is capped at three rounds per agent, which was sufficient to reach convergence in all scenarios. Each experiment is executed three times with different random seeds, and results are averaged to mitigate non-determinism.

C. Baselines and Comparison Methods

We compare against four baselines spanning increasing levels of multi-agent coordination.

Single-agent baseline. A single LLM instance performs RE in a single pass, without role specialization or iterative refinement.

Multi-agent without negotiation. Five specialized agents operate in parallel (corresponding to Phase 1 of our pipeline), with outputs directly aggregated without conflict resolution. This ablation isolates the effect of dialectical negotiation (RQ2).

MARE. A task-specialized multi-agent framework with five agents organized around a 9-action pipeline. Agents collaborate through a shared workspace with single-turn negotiation, decomposing RE by *task type* rather than by quality dimension.

iReDev. A knowledge-driven multi-agent framework with six agents and a 17-action workflow. In our reimplementaion, human-in-the-loop interactions are operationalized via LLM-based surrogate agents with deterministic fallback, enabling full automation and reproducibility while acknowledging that this may not fully capture the quality of real human feedback.

The three multi-agent frameworks differ fundamentally in agent decomposition strategy: MARE assigns agents by *engineering task*, iReDev by *knowledge role*, and QUARE by *quality dimension* (ISO/IEC 25010). This distinction enables evaluation of both absolute performance and the impact of decomposition philosophy on requirement quality, diversity, and compliance.

Since neither MARE nor iReDev provides publicly available implementations, we reimplemented both following their published descriptions. All three frameworks were evaluated under

identical conditions: gpt-4o-mini at temperature 0.7, round cap 3, max tokens 4000, the same five case studies (Table II), four experimental settings, and three random seeds (101, 202, 303), yielding 180 total runs (60 per system). All frameworks are co-implemented within the OpenReBench suite. Where implementation details were underspecified, we adopted the most conservative interpretation consistent with each paper’s stated design principles.

D. Evaluation Metrics

We define evaluation metrics aligned with our three RQs, covering requirement diversity, negotiation effectiveness, and structural validity under regulatory constraints.

a) *RQ1: Coverage and Diversity*: This RQ examines whether the generated requirements span diverse quality concerns, rather than clustering around a small set of dominant attributes. To this end, we embed requirements into a five-dimensional quality space $\mathcal{Q} = \mathbb{R}^5$, where each axis corresponds to one of the five quality attributes managed by the specialized agents: Safety (q_1), Efficiency (q_2), Sustainability (q_3), Trustworthiness (q_4), and Responsibility (q_5).

Each requirement r_i is projected into $\mathcal{Q}$ by scoring its relevance to each quality axis on a $[0, 1]$ scale, using a zero-shot LLM classifier (gpt-4o-mini) prompted with axis-specific rubrics (e.g., the Safety rubric: ‘Rate how directly this requirement addresses hazard identification, fault tolerance, or safety-critical behavior’). All five axes are evaluated in a single call, producing a five-dimensional relevance vector. The classifier is identical to the one used in the blind re-labeling analysis in Section V. Full rubrics are provided in the replication package. Let $P = \{r_1, \dots, r_m\} \subset \mathcal{Q}$ denote the resulting set of projected vectors, treated as a point set in the geometric sense for the coverage measures below.

We quantify coverage and diversity using two geometry-based measures. The first is the convex hull volume (CHV), defined as the volume of $\text{conv}(P)$ in $\mathbb{R}^5$, computed via standard Delaunay triangulation. Higher CHV values indicate broader coverage across quality dimensions. The second metric is the mean distance to centroid (MDC), which captures the dispersion of requirements within the quality space. Formally, let $\bar{q} = \frac{1}{m} \sum_{i=1}^m r_i$ denote the centroid of P , and define

$$\text{MDC} = \frac{1}{m} \sum_{i=1}^m \|r_i - \bar{q}\|_2. \quad (3)$$

A higher MDC reflects greater dispersion and reduced redundancy among generated requirements.

To complement the geometric measures above, which capture overall spread in quality space, we introduce two per-axis distribution metrics that characterize dimensional balance. For multi-agent frameworks whose architecture decomposes requirements by quality dimension, let n_k ($k = 1, \dots, 5$) denote the number of requirements assigned to axis k , and

let $\bar{n} = \frac{1}{5} \sum_k n_k$. Coverage Uniformity (CU) is defined as the standard deviation of per-axis counts:

$$\text{CU} = \sqrt{\frac{1}{5} \sum_{k=1}^5 (n_k - \bar{n})^2}. \quad (4)$$

Lower CU indicates more balanced coverage across quality dimensions. Minimum Axis Coverage (MAC) is defined as $\text{MAC} = \min_k n_k$, capturing the worst-case dimensional guarantee. Higher MAC ensures that no single quality dimension is systematically neglected. CU and MAC require per-axis assignment of requirements to quality dimensions. Multi-agent frameworks provide this naturally through agent specialization, whereas the single-agent baseline lacks such decomposition, we therefore omit CU and MAC for the single-agent baseline.

b) *RQ2: Negotiation Effectiveness*: This RQ evaluates whether dialectical negotiation successfully resolves conflicts while preserving original stakeholder intent. We measure negotiation effectiveness along three complementary dimensions. First, the conflict resolution rate (CRR) measures the proportion of detected contradictions that negotiation successfully resolves. Second, we assess semantic preservation using BERTScore. Finally, logical consistency (S_{logic}) is evaluated through ISO 29148-based checks, detecting numeric or boolean inconsistencies across the resulting requirement set.

c) *RQ3: Structural Validity and Compliance*: This RQ assesses whether negotiated requirements can be reliably mapped to formal goal models and validated against regulatory standards. Structural validity is evaluated by verifying acyclicity and parent-child integrity in the resulting KAOS goal graph. Compliance coverage measures the proportion of applicable regulatory clauses (e.g., ISO 26262) satisfied by the generated model. In addition, terminology consistency (S_{term}) is assessed on a 1–5 scale following ISO 29148, capturing the semantic coherence of concepts across the model and ensuring unified naming (e.g., harmonizing “client” and “user”).

E. Compliance Coverage via RAG-based Clause Entailment

To evaluate industry alignment, we measure *compliance coverage* against domain standards (e.g., ISO 26262 for AD; ISO 27001 for security-centric systems). Let $\mathcal{C}$ denote a corpus of standard clauses. For each case study, we first apply a deterministic scope filter (domain keyword rules) to $\mathcal{C}$, then pass the retained clauses through an LLM-based applicability classifier that outputs a binary label with a short justification, yielding an applicable clause set $\mathcal{C}_{\text{app}} \subseteq \mathcal{C}$. Both the classifier prompt and the model are shared across all compared frameworks.

For each applicable clause $c \in \mathcal{C}_{\text{app}}$, we retrieve the top- k relevant requirements $\{r_1, \dots, r_k\} \subseteq \mathcal{R}$ via embedding similarity, along with supporting standard context (neighboring paragraphs, definitions) to form an evidence bundle $E(c) = \{\text{clause text, context, } r_1, \dots, r_k\}$. An independent verifier LLM then predicts an entailment label $y(c) \in \{\text{SATISFIED, PARTIALLY, NOTSATISFIED}\}$, together with the best supporting requirement, a short rationale, and an extractive citation. A clause is counted as *covered* when $y(c) \in$

Algorithm 1 Set-level semantic preservation via optimal matching

Require: Requirement sets $\mathcal{R}^{(a)} = \{r_i^{(a)}\}_{i=1}^m$, $\mathcal{R}^{(b)} = \{r_j^{(b)}\}_{j=1}^n$
Ensure: Preservation score $P \in [0, 1]$
1: Compute pairwise similarity $S_{ij} \leftarrow \text{BERTScoreF1}(r_i^{(a)}, r_j^{(b)})$
2: $N \leftarrow \max(m, n)$
3: Pad S to $S' \in \mathbb{R}^{N \times N}$ with dummy rows/cols; set dummy similarities to 0
4: Solve $\pi^* \leftarrow \text{HungarianMax}(S') \triangleright \text{max-sum assignment}$
5: $P \leftarrow \frac{1}{N} \sum_{i=1}^N S'_{i, \pi^*(i)}$
6: **return** P

$\{\text{SATISFIED}, \text{PARTIALLY}\}$. Compliance coverage is defined as the proportion of applicable clauses that are covered:

$$\text{Coverage}(\mathcal{R}) = \frac{1}{|\mathcal{C}_{\text{app}}|} \sum_{c \in \mathcal{C}_{\text{app}}} \mathbf{1}[y(c) \in \{\text{SATISFIED}, \text{PARTIALLY}\}] \quad (5)$$

We report coverage averaged over three random seeds. To reduce verifier variance, we fix the verifier model and decoding parameters and apply a self-consistency majority vote over $n=3$ repeated judgments per clause.

F. Set-level Semantic Preservation via Optimal Bipartite Matching

We measure semantic preservation across pipeline phases using a set-level extension of BERTScore. Let $\mathcal{R}^{(a)} = \{r_1^{(a)}, \dots, r_m^{(a)}\}$ and $\mathcal{R}^{(b)} = \{r_1^{(b)}, \dots, r_n^{(b)}\}$ denote the requirement sets produced at phases a and b (e.g., Phase 1 vs. Phase 3). We compute a pairwise similarity matrix $S_{ij} = \text{BERTScoreF1}(r_i^{(a)}, r_j^{(b)})$ using bert-base-uncased, then solve a maximum-weight bipartite matching to obtain a one-to-one alignment that accounts for set-size differences via dummy padding. Algorithm 1 summarizes the procedure, the resulting score $P \in [0, 1]$ decreases when requirements are rewritten (lower pairwise similarity) or when the set size changes (unmatched dummy nodes contribute zero similarity).

V. RESULTS AND ANALYSIS

In this section, we present results validating the effectiveness of QUARE. Our analysis follows three RQs, demonstrating how the framework progresses from diverse generation to consistent, engineering-ready models.

A. Results for RQ1: Coverage and Diversity

RQ1 examines whether specialized multi-agent collaboration improves the coverage and diversity of generated requirements relative to single-agent generation. We compare the single-agent baseline with Phase 1 outputs, in which five quality-oriented agents independently generate requirements without negotiation. All geometric metrics (CHV and MDC) are computed in the original five-dimensional quality space $\mathcal{Q} = \mathbb{R}^5$, with PCA used solely for visualization.

TABLE III: Requirement Coverage and Diversity (averaged across 5 case studies)

Framework	Req. Count ($\uparrow$)	CHV ($\uparrow$)	MDC ($\uparrow$)	CU ($\sigma, \downarrow$)	MAC ($\uparrow$)
Single-agent	14.2	2.8	0.715	—	—
MARE	24.4	4.8	0.835	0.30	4.6
iReDev	28.1	6.4	0.705	0.45	5.1
QUARE	35.0	4.3	0.673	0.20	6.7

Req. Count: total number of unique generated requirements. **CHV:** convex hull volume in 5-D quality space (values $\times 10^{-3}$). **MDC:** mean distance to centroid. **CU:** coverage uniformity (Eq. 4); lower is more balanced. **MAC:** minimum axis coverage, i.e., the lowest per-axis requirement count; higher values indicate stronger worst-case coverage. CU and MAC are not applicable to the single-agent baseline, which lacks quality-dimension decomposition.

The quantitative results in Table III confirm that multi-agent outputs achieve broader coverage than single-agent baselines across all five case studies. We provide detailed per-case breakdowns and visualizations in the supplementary material. The three multi-agent systems exhibit complementary strengths across coverage metrics, reflecting distinct architectural philosophies. QUARE generates the most requirements (35.0 avg, +43% over MARE, +25% over iReDev), driven by five quality-specialized agents that systematically surface dimension-specific concerns. iReDev achieves the highest CHV (0.0064, +48% over QUARE), indicating the broadest quality-space exploration through its 6-agent knowledge-driven pipeline. MARE leads MDC (0.835, +24% over QUARE), producing the most dispersed, non-redundant distribution.

While QUARE’s CHV and MDC are lower than those of iReDev and MARE respectively, this reflects an architectural trade-off inherent to quality-specialized decomposition. Because each agent focuses on a single quality dimension, generated requirements naturally cluster along their assigned axes, reducing geometric spread but improving dimensional balance and worst-case coverage. QUARE achieves the lowest CU (0.20 vs. 0.30 for MARE and 0.45 for iReDev) and the highest MAC (6.7 vs. 4.6 and 5.1), meaning that even its least-represented dimension receives more requirements than the worst-case axis of either baseline. This balanced coverage directly supports compliance outcomes: in the AD case, the Safety agent’s requirements achieve 91.1% ISO 26262 coverage versus MARE’s 7.8% and iReDev’s 10.0%, indicating that domain-specific emphasis is expressed through semantic depth rather than raw count. These results reveal an inherent trade-off between architectural design and coverage characteristics. Task-specialized agents (MARE) maximize geometric dispersion, knowledge-driven agents (iReDev) maximize coverage breadth, and quality-specialized agents (QUARE) ensure balanced per-axis coverage with the strongest worst-case dimensional guarantees, a property that translates into downstream compliance outcomes (Table VI).

Answer to RQ1

Multi-agent collaboration yields up to $2.5\times$ more requirements, 129% higher CHV, and 17% higher MDC

TABLE IV: Semantic Preservation (BERTScore F1, Phase 3 vs. Phase 1, $\uparrow$)

Case Study	MARE	iReDev	QUARE
AD	88.4	92.7	94.8
ATM	89.6	92.0	95.5
Library	87.7	92.6	94.8
RollCall	88.8	93.0	94.5
Bookkeeping	90.5	92.9	94.8
Average	89.0	92.6	94.9

compared to single-agent generation. Among multi-agent frameworks, each architectural philosophy produces distinct coverage characteristics: QUARE generates the most requirements (35.0, +43% over MARE) with the most balanced dimensional coverage (CU=0.20 vs. 0.30/0.45) and strongest worst-case axis guarantee (MAC=6.7 vs. 4.6/5.1); iReDev achieves the broadest quality-space exploration (CHV=0.0064); and MARE produces the most dispersed distribution (MDC=0.835). QUARE’s balanced coverage translates into compliance advantages examined in RQ3 (98.2% vs. ~47%).

B. Results for RQ2: Negotiation Effectiveness and Semantic Preservation

RQ2 investigates how dialectical negotiation resolves conflicts while preserving semantic intent. We perform a phase-wise trajectory analysis across four milestones: Phase 1 (Raw Generation), Phase 2 (Negotiated Alignment), Phase 3 (Unified Integration), and Phase 4 (Verified Output), comparing all three frameworks under identical conditions.

1) *Quantitative Results*: Table IV shows that QUARE achieves the highest semantic preservation across all cases (94.9% avg BERTScore between Phase 3 and Phase 1 outputs, +5.9 percentage points over MARE, +2.3 percentage points over iReDev), with perfect Phase 2 vs. Phase 1 preservation (100.0%). Table V further shows that this result is achieved despite conducting the deepest negotiation among all three systems (16.5 steps avg).

The lower CRR of QUARE (25.0% vs. MARE’s 66.7%) reflects a difference in detection granularity rather than resolution quality. MARE employs coarser conflict detection, marking fewer total conflicts and resolving most through simple priority override. QUARE’s two-stage detection (embedding similarity + LLM classification) surfaces more conflicts, including architectural inconsistencies and cross-level semantic misalignments that coarser detection would miss. The larger denominator lowers the resolution rate. Unresolved conflicts are not discarded but handled through priority-weighted integration in Phase 3, as reflected in QUARE’s higher overall semantic preservation (Table IV). Maximizing CRR is therefore less important than surfacing meaningful conflicts and preserving intent across the pipeline.

2) *Case Study: Multi-Agent Negotiation in Autonomous Driving*: To illustrate the dialogical protocol concretely, we examine the sensor fusion latency conflict in the AD system, one of the highest-severity incompatibilities detected across all

TABLE V: Negotiation Process Summary

Metric	MARE	iReDev	QUARE
Avg. negotiation steps	10.0	12.0	16.5
Conflict resolution rate (%)	66.7	46.7	25.0*
Phase 2 vs. Phase 1 BERTScore (%)	97.0	99.4	100.0

*QUARE detects 3.2 $\times$ more conflicts than MARE; unresolved conflicts are handled via priority-weighted integration in Phase 3 (see Section III-C).

five case studies. During Phase 1, the SafetyAgent proposed sensor fusion latency ≤ 500 ms (for dual-channel verification per ISO 26262-5 ASIL D compliance), while the EfficiencyAgent mandated ≤ 30 ms (for 10 Hz real-time planning cycles per Apollo Spec §4.2), a 16.7 $\times$ numeric gap that neither MARE’s nor iReDev’s implicit aggregation mechanisms could structurally resolve.

Round 1: Conflict Eruption

SafetyAgent (TG2): Sensor fusion latency ≤ 500 ms for comprehensive sensor validation and fault detection per ISO 26262-5.

EfficiencyAgent: *Strong opposition*. 500 ms consumes 5 $\times$ the planning cycle budget (100 ms target). Sensor fusion must complete within 30 ms per Apollo Spec §4.2.

Classification: High-severity numeric incompatibility (resource-bound). **UNRESOLVED**

Round 2: Architectural Refinement

SafetyAgent (revision): Two-stage pipeline—fast-path ≤ 30 ms (95% cases), thorough-path ≤ 500 ms (anomaly-triggered).

EfficiencyAgent: *Insufficient*. Thorough-path must be (1) asynchronous and non-blocking, (2) fast-path $\geq 99\%$, (3) fallback ensures planning continuity. **PARTIAL**

Round 3: Consensus & Convergence

SafetyAgent (final): Three-component architecture:

- TG2.1: Fast-path ≤ 30 ms, synchronous, $\geq 99\%$
- TG2.2: Thorough-path ≤ 500 ms, async, non-blocking
- TG2.3: Fallback—planning uses fast-path during validation

EfficiencyAgent: Fast-path satisfies G1. Async thorough-path has zero planning impact. **CONSENSUS**

The subsequent negotiation proceeded through three rounds of structured dialogical refinement, transforming this trade-off into an architectural solution satisfying both agents’ core constraints. This negotiation trace demonstrates how dialogical negotiation enables *creative architectural solutions* rather than simple compromises or priority-based rejection. By decomposing the conflicting requirement into a synchronous fast-path (efficiency-critical, ≤ 30 ms, $\geq 99\%$ of cases) and an asynchronous thorough-path (safety-critical, ≤ 500 ms, non-blocking), the framework satisfies both agents’ constraints without sacrificing either safety or efficiency. This architectural decomposition pattern reflects the broader negotiation behavior exhibited by QUARE across all five case studies, in which conflicts are resolved by restructuring the solution space rather than conceding individual constraints.

In contrast, MARE would likely resolve this via safety priority override, discarding the efficiency constraint, while iReDev lacks the structured dialogical mechanism to reach the three-

component synthesis. This pattern generalizes across all five case studies, the protocol resolves conflicts by restructuring the solution space rather than conceding individual constraints.

Answer to RQ2

QUARE achieves 94.9% semantic preservation (+5.9 percentage points over MARE, +2.3 percentage points over iReDev) despite conducting the deepest negotiation among all three systems (16.5 steps avg). The dialogical protocol resolves cross-agent conflicts through architectural decomposition rather than priority override, producing requirements that are both logically coherent and engineering-ready.

C. Results for RQ3: Structural Validity and Compliance

RQ3 examines whether QUARE generates structurally valid and industry-compliant requirement models. We evaluate outputs from Phase 3 (model integration) and Phase 4 (verification and compliance checking), comparing all three frameworks under identical experimental conditions.

1) *Structural Validity and Internal Consistency*: As shown in Table VI (A), all three systems achieve comparable structural validity: valid DAG topology and zero detected logical contradictions ($S_{\text{logic}} = 1.000$) across all case studies. Each operational element traces to at least one tactical goal, and each tactical goal links to a strategic objective. The key differentiator is compliance coverage. QUARE achieves 98.2% average compliance, a +105% improvement over both MARE (47.6%) and iReDev (47.8%), which score nearly identically despite differing architectures. This gap is most pronounced in the safety-critical AD case, where QUARE reaches 91.1% ISO 26262 coverage against MARE’s 7.8% and iReDev’s 10.0%. This advantage stems from Phase 4 RAG-augmented verification, which maps requirements to specific standard clauses. Neither MARE nor iReDev includes such a step; their compliance scores reflect only incidental overlap between generated requirements and regulatory clauses.

TABLE VI: Structural Validity and Compliance

(A) Structural Correctness			
Metric	MARE	iReDev	QUARE
DAG Topology Valid	✓	✓	✓
Logical Consistency (S_{logic})	1.000	1.000	1.000
Compliance Coverage (%)	47.6	47.8	98.2

(B) ISO/IEC/IEEE 29148 Quality Scores (1–5 scale, LLM-judge)

Criterion	MARE	iReDev	QUARE
Unambiguous	4.41	4.19	4.24
Correctness	5.00	5.00	5.00
Verifiability	3.95	3.96	4.96
Set Consistency	5.00	5.00	5.00
Set Feasibility	3.74	3.75	4.96

Note: Compliance coverage averages ISO 26262 (AD) and ISO 27001 (remaining cases). Bold indicates the best result per metric.

2) *ISO/IEC/IEEE 29148 Evaluation*: We evaluate industrial quality against ISO 29148 at both individual requirement

and requirement-set levels, using an LLM-based judge instantiated independently of the generation agents to mitigate self-evaluation bias. Table VI (B) presents cross-system averages. All three systems score 5.00 on both correctness and consistency, indicating that multi-agent pipelines reliably produce well-formed, internally coherent requirement sets. MARE leads marginally in unambiguity (4.41), which we attribute to its task-specialized agents’ focus on specification-level phrasing. The main advantages of QUARE appear in verifiability (4.96, +25% over MARE) and set feasibility (4.96, +32% over MARE). Both gains trace to the RAG-augmented verification phase, which enforces measurable acceptance criteria and checks implementation feasibility against domain standards. These gains hold across all five case studies and are the quality dimensions most relevant to industrial adoption.

3) *Regulatory Compliance and Semantic Stability*: Phase 4 applies deterministic rule checking and standards-based validation (ISO 26262 for AD, ISO 27001 for security systems). After verification, all QUARE-generated models achieve 100% compliance. The BERTScore between Phase 3 and Phase 4 remains unchanged across all cases, confirming that compliance checking functions as a pure validation layer rather than a transformation step, and that negotiated semantic intent is fully preserved throughout the verification pipeline.

Answer to RQ3

All three systems achieve structural parity (valid DAG topology, $S_{\text{logic}} = 1.000$, perfect correctness and consistency). However, **QUARE** outperforms both MARE and iReDev in compliance coverage (**98.2%** vs. 47.6% and 47.8%) and requirement quality (verifiability **4.96**, feasibility **4.96**), demonstrating that automated RAG-augmented verification is the key component for producing engineering-ready, standards-compliant artifacts.

D. Cross-Framework Synthesis

Synthesizing the results from Tables III–VI, QUARE leads in compliance-critical metrics (compliance coverage, verifiability, feasibility, and semantic preservation), requirement count, and dimensional balance (lowest CU, highest MAC), while MARE leads in runtime and dispersion (MDC), and iReDev achieves the broadest quality-space exploration (CHV). The three systems reflect distinct design philosophies: QUARE excels in compliance-critical and safety-critical scenarios with the most balanced quality-dimension coverage, strongest semantic preservation, and regulatory alignment at moderate cost (55.4s); MARE is optimal for rapid prototyping with the fastest runtime, but lower compliance; and iReDev occupies a knowledge-intensive middle ground with the highest CHV but at $4.8\times$ computational cost of MARE.

E. Ablation Study and Discussion

This section provides a consolidated analysis of how individual components of QUARE contribute to the overall performance observed in RQ1–RQ3. Our ablation follows a

phase-wise design that incrementally evaluates the effect of each processing stage.

Agent Specialization (Phase 1): CHV increases 53.6% over single-agent baselines, confirming that diversity emerges from quality-oriented specialization. MDC decreases modestly (5.9%), consistent with the architectural trade-off discussed in Section V-A: quality-specialized agents produce requirements that cluster along their assigned axes, yielding lower geometric dispersion but stronger dimensional balance ($CU = 0.20$, $MAC = 6.7$).

Explicit Negotiation (Phase 2): QUARE achieves perfect semantic retention during negotiation (P2 vs. P1: 100.0%), indicating that the dialogical protocol resolves conflicts without overwriting original stakeholder intent. MARE and iReDev show slight drift (97.0% and 99.4%, respectively).

Model Integration (Phase 3): Integration introduces modest semantic adjustment (P3 vs. P1: 94.9% for QUARE), primarily from deduplication and structural harmonization rather than intent modification.

Verification and Compliance Checking (Phase 4): Zero semantic drift (P4 vs. P3 unchanged), confirming that compliance checking functions as a pure validation layer.

The design of QUARE offers several properties relevant to industrial adoption. First, the quality-specialized decomposition is extensible to additional agents without restructuring the pipeline. Second, the negotiation trace provides an auditable record of trade-off decisions, addressing traceability requirements in safety-critical domains (ISO 26262, IEC 62304) and emerging regulatory mandates (EU AI Act, Article 11). Third, the moderate runtime (55.4s) and compatibility with lightweight models suggest feasibility for iterative development workflows.

VI. THREATS TO VALIDITY

Aligning with established guidelines for empirical software engineering research, we discuss potential threats to the validity of our study. These threats are organized into four categories: construct validity, internal validity, conclusion validity, and external validity.

A. Construct Validity

The CHV and MDC metrics rely on LLM-based quality classification. Blind re-labeling with an independent classifier yields 75% agreement and stable metric values, suggesting robustness. BERTScore is complemented by the logical consistency score (S_{logic}) to capture inconsistencies beyond embedding similarity. The ISO 29148 quality assessment also relies on an LLM-based judge, which may exhibit systematic biases. We mitigate this through axis-specific rubrics and independent blind relabeling (75% agreement).

B. Internal Validity

Since neither MARE nor iReDev provides publicly available implementations, both were reimplemented based on their published descriptions within the shared OpenReBench framework

under identical conditions. Where the original specification was ambiguous, we adopted conservative design decisions.

Improvements may be driven by increased computational effort rather than architecture. We control model, prompt budget, and environment across all baselines, and isolate contributions via phase-wise ablation. All experiments use three random seeds with averaged results to reduce stochastic bias.

C. External Validity

The evaluation covers five case studies across safety-critical, financial, and information-management domains, though this may not capture the full range of RE scenarios. The AD case was constructed from public industrial documentation. As reimplementations may not perfectly replicate original behavior, results on MARE and iReDev should be interpreted as reflecting best-effort reconstruction. Reliance on a single LLM (gpt-4o-mini) limits generalizability.

D. Conclusion Validity

Results are averaged over three random seeds (101, 202, 303), with consistent ordering across all five case studies. The absence of formal significance testing (e.g., Wilcoxon signed-rank test) is a limitation, constrained by the computational cost of 180 total LLM runs. Future work will incorporate human expert evaluation, nonparametric significance tests, additional repetitions, and validation with alternative models.

VII. CONCLUSION

This paper presented QUARE, a structured multi-agent framework that decomposes requirements analysis into quality-specialized agents, resolves cross-quality conflicts through explicit dialogical negotiation, and produces engineering-ready KAOS goal models with automated compliance verification. Through a three-way evaluation against MARE and iReDev across 180 experiment runs, we showed that (1) agent specialization yields up to $2.5\times$ more requirements than single-agent generation with balanced quality-axis coverage ($CU=0.20$), while each architectural approach offers distinct diversity characteristics (RQ1), (2) explicit negotiation achieves the highest semantic preservation (94.9%) while resolving conflicts through architectural decomposition rather than priority override (RQ2), and (3) all frameworks achieve structural validity, while RAG-augmented verification further yields 98.2% compliance coverage, a +105% improvement over both baselines (RQ3).

For researchers, these findings suggest that effective RE automation depends more on principled architectural decomposition and explicit interaction protocols than on raw model scale. For practitioners, the modular design of QUARE is compatible with lightweight models, its quality-specialized agents map to organizational roles, and its negotiation traces can support regulatory audits.

As future work, we plan to explore tighter human-in-the-loop integration and apply QUARE to larger-scale industrial settings. In addition, while the current negotiation protocol

draws on dialogical principles, the formal argumentation literature offers conflict resolution mechanisms such as value-based argumentation frameworks and abstract argumentation semantics, both largely unexplored in RE. Grounding the negotiation protocol in these theories could yield stronger formal guarantees on completeness and admissibility.

DATA AVAILABILITY STATEMENT

All experimental artifacts supporting the findings of this study, including framework implementations of QUARE, MARE, and iReDev, prompt templates, case study inputs, raw outputs, and evaluation scripts, are openly available at: <https://anonymous.4open.science/t/OpenRE-Bench-6660>.

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